Given a vector $\overrightarrow{AB}$, the **length of the vector** is calculated by

$$\sqrt{(b_1-a_1)^2+(b_2-a_2)^2+(b_3-a_3)^2},$$

where A is (a_1,a_2,a_3) and B is (b_1,b_2,b_3). In particular, if a vector is the position vector $\vec{b}$ of (b_1,b_2,b_3), then its length is $\sqrt{b_1^2+b_2^2+b_3^2}$. A vector having length 1 is called a **unit vector**. We use the notation $\hat{u}$, for a unit vector. Note that $\hat{i},\hat{j}$, and $\hat{k}$ are unit vectors and $\vec{0}$ is the unique vector with length 0. The direction of $\vec{0}$ is specified according to the context.

The **addition** and **scalar multiplication** on vectors in 3-dimensional space are defined by

$$\vec{a}+\vec{b} = (a_1+b_1)\hat{i}+(a_2+b_2)\hat{j}+(a_3+b_3)\hat{k}.$$

$$\alpha\vec{a} = (\alpha a_1)\hat{i}+(\alpha a_2)\hat{j}+(\alpha a_3)\hat{k};$$

where $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k},\ \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}\in\mathbb{R}^3$ and $\alpha\in\mathbb{R}$.

To see the geometric interpretation of $\vec{a}+\vec{b}$, let $\vec{a}$ and $\vec{b}$, denote the position vectors of $A=(a_1,a_2,a_3)$ and $B=(b_1,b_2,b_3)$, respectively. Translate the position vector $\vec{b}$ to the vector with initial point as A and end point as $C=(c_1,c_2,c_3)$, for a suitable $(c_1,c_2,c_3)\in\mathbb{R}^3$. See the Fig (6.2). Then, the position vector $\vec{c}$ of the point (c_1,c_2,c_3) is equal to $\vec{a}+\vec{b}$.

The vector $\alpha\vec{a}$ is another vector parallel to $\vec{a}$ and its length is magnified (if $\alpha>1$) or contracted (if $0<\alpha<1$). If $\alpha<0$, then $\alpha\vec{a}$ is a vector whose magnitude is $|\alpha|$ times that of $\vec{a}$ and direction opposite to that of $\vec{a}$. In particular, if $\alpha=-1$, then $\alpha\vec{a}=-\vec{a}$ is the vector with same length and direction opposite to that of $\vec{a}$. See Fig. 6.3

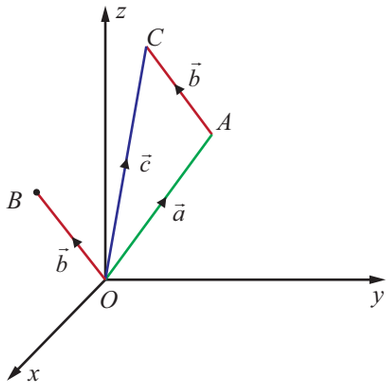


Fig. 6.2

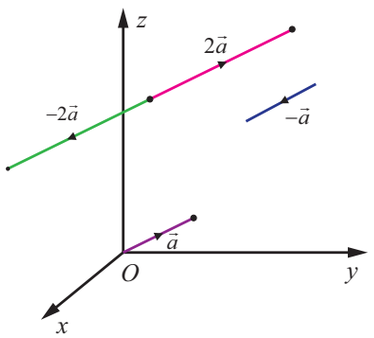


Fig. 6.3

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